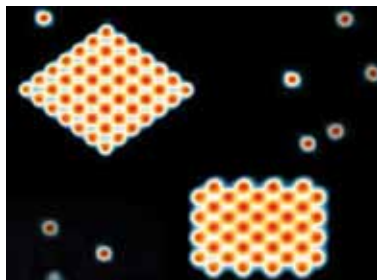




Atom-by-atom construction of new superconductors

Researchers at the University of Zurich have achieved a remarkable feat by meticulously designing superconductors, atom by atom, leading to the creation of previously



Scanning tunneling microscope image of two of the superconducting structures created, which consist of individual chromium atoms (Credit: University of Zurich)

unexplored states of matter. Researchers created materials atom by atom, focusing on pioneering superconductors that exhibit zero electrical resistance at low temperatures. Often called ‘ideal diamagnets’ these have extensive use in quantum computers for their unique magnetic interactions. Using a scanning tunnelling microscope, researchers manipulated atoms with precision, measuring the system’s magnetic and superconductive characteristics. By adding chromium atoms to superconducting niobium, they created two new superconductivity forms. This approach marks the first successful creation of two-dimensional superconductors, although similar methods have been used before with metal atoms.

A device generates hydrogen from sunlight

Rice University engineers achieved unprecedented efficiency in converting sunlight into hydrogen by integrating cutting-edge halide perovskite semiconductors and electrocatalysts into a single, durable, cost-effective, and easily scalable unit. The researchers emphasised that using sunlight for chemical production is a significant challenge.



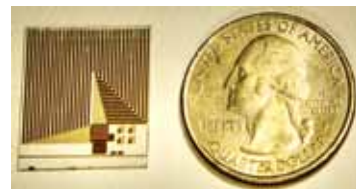
Series of four still images from a sample video showing how a photoreactor from Rice University splits water molecules and generates hydrogen when stimulated by simulated sunlight (Credit: Mohite lab/Rice University)

Their solution involves designing a system to absorb light and conduct electrochemical water-splitting chemistry on its surface. The team transformed their efficient solar cell into a reactor, enabling the harvested energy to split water into oxygen and hydrogen. The researchers identified the need for a dual-layer barrier, with one layer to prevent

water infiltration and another to establish optimal electrical contact between the perovskite layers and the protective coating. Their research yielded the highest efficiency among photoelectrochemical cells without solar concentration and stood out as the overall best for devices utilising halide perovskite semiconductors.

Cell-mimicking sensor replicates membrane functions

Researchers at MIT have created an innovative sensor capable of detecting molecules similar to those identified by naturally occurring cell receptors. The team used water-soluble receptor proteins and bacterial S-layer proteins to create a coherent array on a surface,



Cell-mimicking sensor (Courtesy of the researchers; Edited by MIT News)

densely immobilising a receptor protein (CXCR4) that binds to target molecules, including those involved in HIV entry and cancer. They named this sensing technology RESENSA (Receptor S-layer Electrical Nano Sensing Array). Researchers used crystallised S-layers on a graphene-based chip to create a dual layer with water-soluble receptor proteins. Target molecule binding alters the chip’s electrical properties, enabling easy quantification and data transmission to devices like smartphones. The team envisions a portable device for at-home tests, providing quick medical insights.

Bio-inspired jellyfish robot

Researchers at the Indian Institute of Technology Indore and Jodhpur have developed a robot inspired by biology that resembles and moves like a jellyfish. To create a jellyfish-mimicking robot, the team sliced 75µm-thick Kapton polyimide sheets into a 25cm diameter shape resembling a jellyfish. They made holes for shape memory alloy (SMA) wires, known to revert to their original shape when heated after deformation. They used nitinol, a common SMA, threading it through the pre-made holes in the structure. An extra layer of polyimide tape secured the wire, and rubber strings linked the tentacles to the robot’s body. The lightweight robot, weighing only



The jellyfish robot developed by the researchers (Credit: Muralidharan et al)